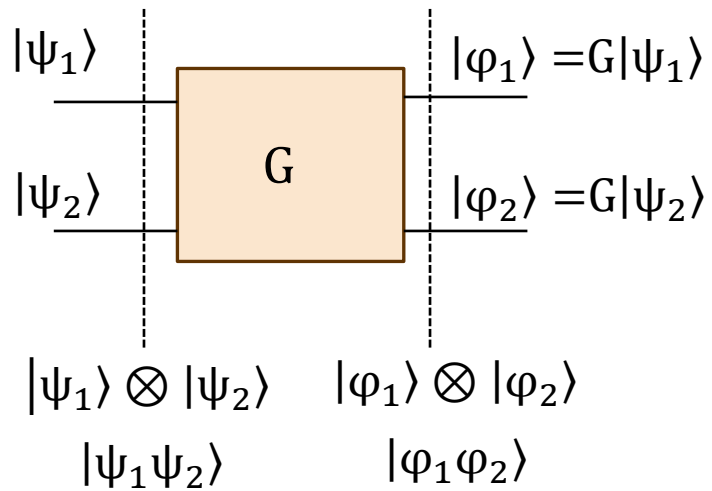


Introduction to Quantum Computing (L-4)

By: Dr. Samik Mukherjee (Ph.D.)
Assistant Professor, Jio Institute

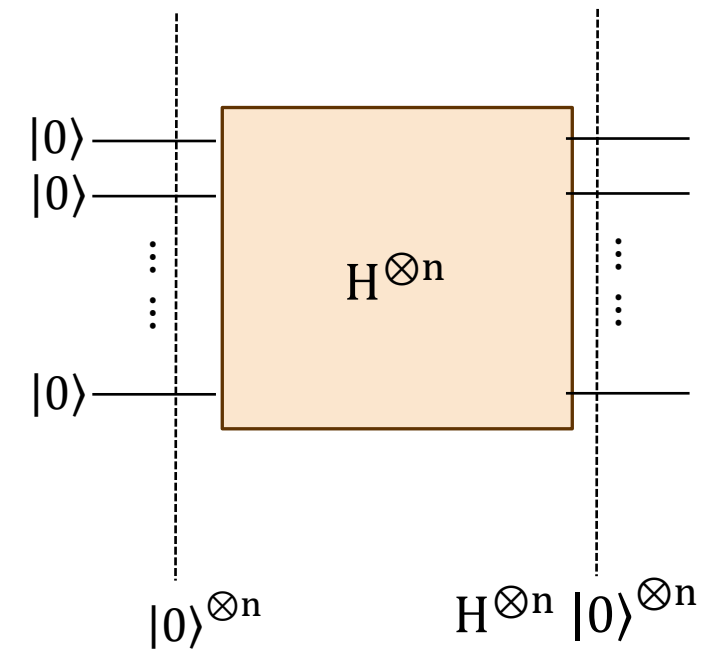
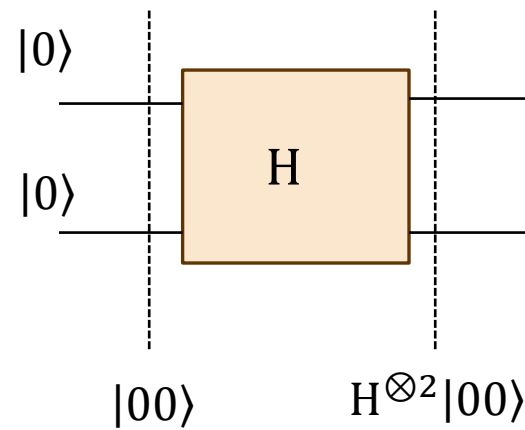
Tensor Products

The tensor product combines individual qubit states into a joint multi-qubit state, e.g. if $|\psi_1\rangle = a|0\rangle + b|1\rangle$ and $|\psi_2\rangle = c|0\rangle + d|1\rangle$, then the two-qubit state is $|\psi\rangle \otimes |\psi_2\rangle$ a 4-dimensional vector describing the full system. For convenience we write this as $|\psi_1\psi_2\rangle$

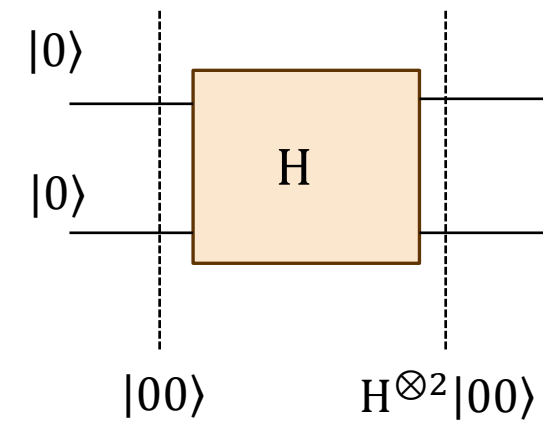


$$\begin{aligned} |\varphi_1\varphi_2\rangle &= |\varphi_1\rangle \otimes |\varphi_2\rangle \\ &= G|\psi_1\rangle \otimes G|\psi_2\rangle \end{aligned}$$

$$G^{\otimes 2}|\psi_1\psi_2\rangle$$



State Vectors



$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}_{2 \times 1}$ $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}_{2 \times 1}$

$|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}_{4 \times 1}$; $|01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}_{4 \times 1}$; $|10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}_{4 \times 1}$; $|11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}_{4 \times 1}$

$|000\rangle = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{8 \times 1}$; $|001\rangle = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}_{8 \times 1}$; $|110\rangle = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}_{8 \times 1}$; $|111\rangle = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}_{8 \times 1}$

A quantum circuit diagram with two horizontal lines representing qubits. Both lines start at the left with the state $|0\rangle$. A vertical dashed line is placed before the first qubit and another after the second. Between these dashed lines is an orange square block labeled $H^{\otimes 2}$. The output of the circuit, after the second dashed line, is labeled $\frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$.

$|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$H^{\otimes 2}|00\rangle = H|0\rangle \otimes H|0\rangle$

$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}$

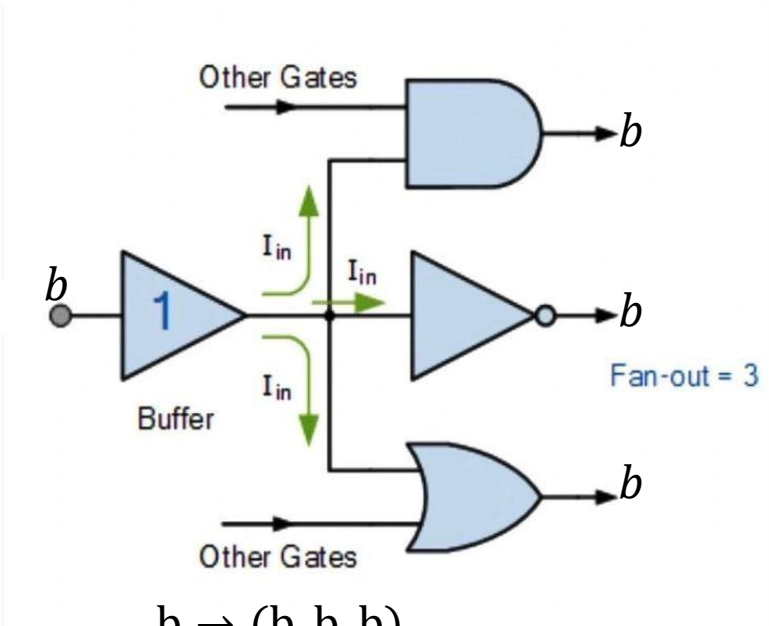
$\frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{\sqrt{2}\sqrt{2}}$

$\frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

No Cloning of an Arbitrary Quantum Bit

Classical fan-out circuits allow us to make as many copies of a classical bit as we want

DIGITAL BUFFER FANOUT



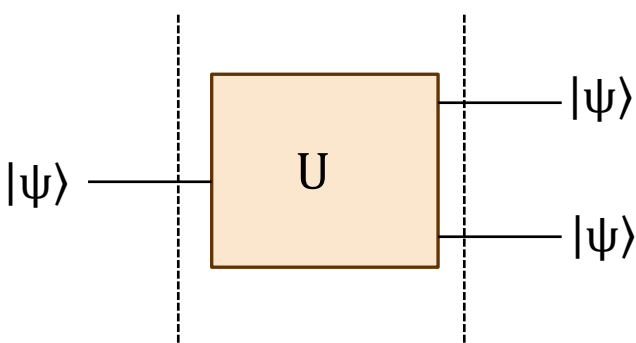
$$b \rightarrow (b, b, b)$$
$$1 \rightarrow (1, 1, 1) \quad 0 \rightarrow (0, 0, 0)$$

- A classical bit has a definite value
- Measuring or copying it does not disturb it
- Classical logic gates are not constrained by linearity or unitarity

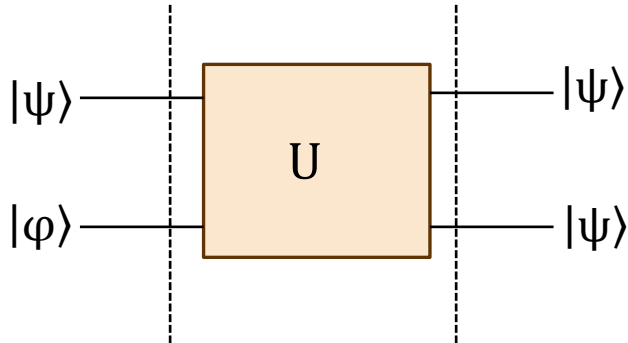
No-Cloning Theorem

It is impossible to make an exact copy of an unknown arbitrary quantum state. Or we can say that there is no Unitary operator (U) that can clone an arbitrary qubit

- Proven in 1982
- Holds for all quantum systems
- A direct consequence of linearity + unitarity nature of quantum gates



Note: This is not a valid Unitary operation. Unitary circuits have the same size of inputs as outputs

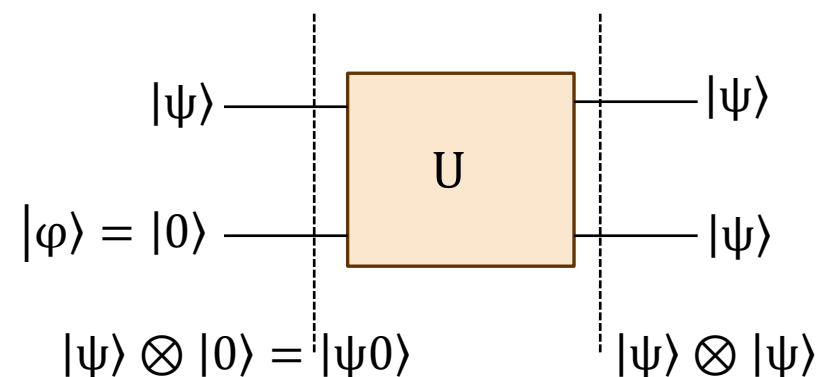


$$U[|\psi\rangle \otimes |\phi\rangle] = |\psi\rangle \otimes |\psi\rangle$$

The above operation sounds reasonable... but **Nature says NO**

No Cloning of an Arbitrary Quantum Bit

Proof by Contradiction: Lets assume that such a unitary circuit exist that can copy arbitrary qubits



Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ be the arbitrary qubit that we want to clone through the linear Unitary operation 'U'

$$|\psi 0\rangle = \alpha|00\rangle + \beta|10\rangle$$

Since U is linear, by linearity property

$$U|\psi 0\rangle = U\{\alpha|00\rangle + \beta|10\rangle\} = \alpha U|00\rangle + \beta U|10\rangle$$

$$U\{\alpha|00\rangle + \beta|10\rangle\} = \alpha U|00\rangle + \beta U|10\rangle$$

LHS gives: $U\{\alpha|00\rangle + \beta|10\rangle\}$ should produce two copies of $\alpha|00\rangle + \beta|10\rangle$ if cloning was possible. That is, at the output we should have $\{\alpha|00\rangle + \beta|10\rangle\} \otimes \{\alpha|00\rangle + \beta|10\rangle\}$

$$\alpha^2|0000\rangle + \alpha\beta|0010\rangle + \beta\alpha|1000\rangle + \beta^2|0010\rangle$$

The RHS $\alpha U|00\rangle + \beta U|10\rangle$ operation will produce $\alpha|00\rangle \otimes |00\rangle + \beta|10\rangle \otimes |10\rangle = \alpha|0000\rangle + \beta|1010\rangle$

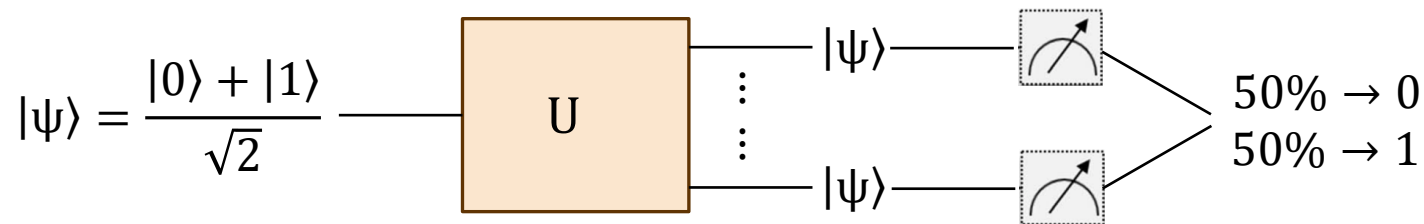
So, if $U\{\alpha|00\rangle + \beta|10\rangle\} = \alpha U|00\rangle + \beta U|10\rangle$ needs to be true (meaning, if we assume that cloning is possible) then

$$\alpha^2 = \alpha \text{ and } \beta^2 = \beta$$

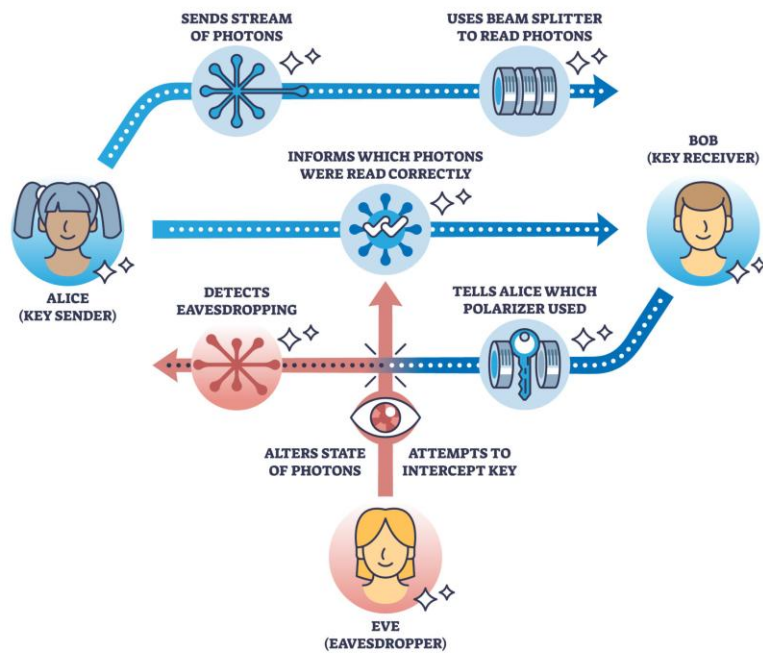
$$\alpha\beta = \beta\alpha = 0 \text{ which can only happen if either } \alpha = 0 \text{ or if } \beta = 0, \text{ or both are } 0$$

The moment we set $\alpha = 0$ or if $\beta = 0$, or both are 0 the equality of LHS and RHS breaks down. This is a proof by contradiction (PEC) that cloning is an arbitrary quantum bit is not allowed.

No Cloning of an Arbitrary Quantum Bit



PROCESS OF QUANTUM KEY DISTRIBUTION (QKD)



If cloning were possible:

- Clone many times
- Measure all copies
- Reconstruct α and β exactly

That would let us:

- Extract full quantum information
- Violate quantum uncertainty
- Break quantum mechanics

Nature prevents this by **forbidding cloning**

Quantum Cryptography (BB84): Because qubits cannot be copied, any eavesdropping attempt necessarily disturbs the state and reveals itself, making security guaranteed by physics rather than computational assumptions.

Quantum Teleportation: The no-cloning principle ensures that the quantum state is destroyed at the sender and recreated at the receiver, preventing any illegal duplication of quantum information.

Quantum Error Correction: Since qubits cannot be backed up by copying, error correction relies on entanglement and carefully distributed redundancy, making it fundamentally different from classical error-correction schemes.

Entanglement

Entanglement is such a state of a multi-qubit system wherein we cannot express the entangled state as a tensor product of individual qubits separately.

Let's say we have a multi-qubit system described by $\frac{|00\rangle + |01\rangle}{\sqrt{2}}$ _____

If the qubits are not entangled (that is, they are separable) then we can always express the multi-qubit state as $(\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$

Let's now take a multi-qubit system described by $\frac{|00\rangle + |11\rangle}{\sqrt{2}}$

Try to see that there is absolutely no combination of $(\alpha, \beta, \gamma, \delta)$ for which we can express this multi-qubit state as $(\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}} = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle$$

$$\alpha\gamma = 1/\sqrt{2} = \beta\delta$$

$\alpha\delta = 0 = \beta\gamma$

- If $\beta = 0$, then $\beta\delta = 1/\sqrt{2}$ will not hold true
- If $\gamma = 0$, then $\alpha\gamma = 1/\sqrt{2}$ will not hold true

$$\frac{|00\rangle + |01\rangle}{\sqrt{2}}$$

$$|0\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

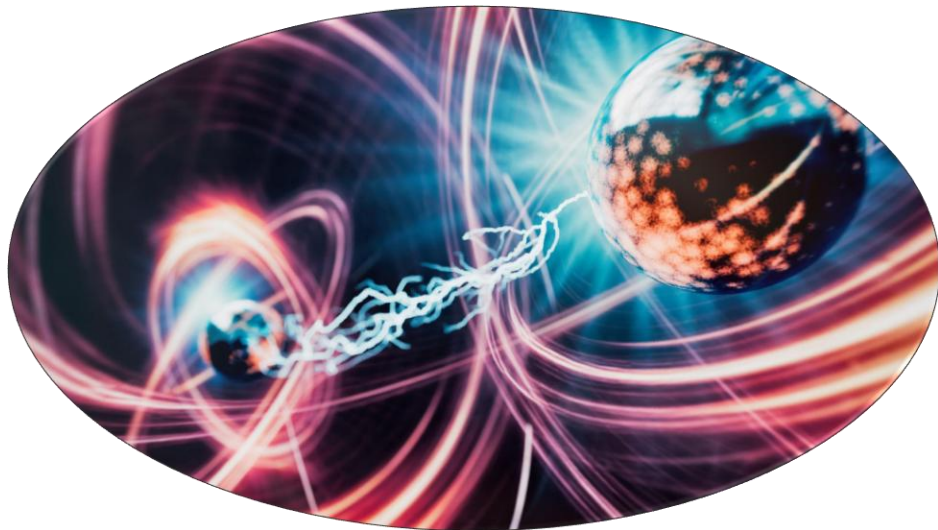
$$(|0\rangle + 0|1\rangle) \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$\alpha = 1; \beta = 0;$$

$$\gamma = 1/\sqrt{2}; \delta = 1/\sqrt{2}$$

This is called entanglement, wherein the state of each qubit cannot be described independently, even though the qubits are spatially separate.

Entanglement



There could be many possible entangled states.

Like, $|\psi\rangle = \sqrt{0.9}|00\rangle + \sqrt{0.1}|11\rangle$ is also an entangled state. Try yourself to see that you cannot express $\sqrt{0.9}|00\rangle + \sqrt{0.1}|11\rangle$ as a tensor product of two individual qubit states.

However, there are a special class of entangled state which are maximally entangled, or rather, the individual qubits participating in the entanglement share the strongest correlation. This happens when $\alpha = \beta = 1/\sqrt{2}$. For any non-equal (α, β) the correlation is only partial

Such maximally entangled states are called Bell states. Sometimes people also call them the Einstein–Podolsky–Rosen or EPR states.

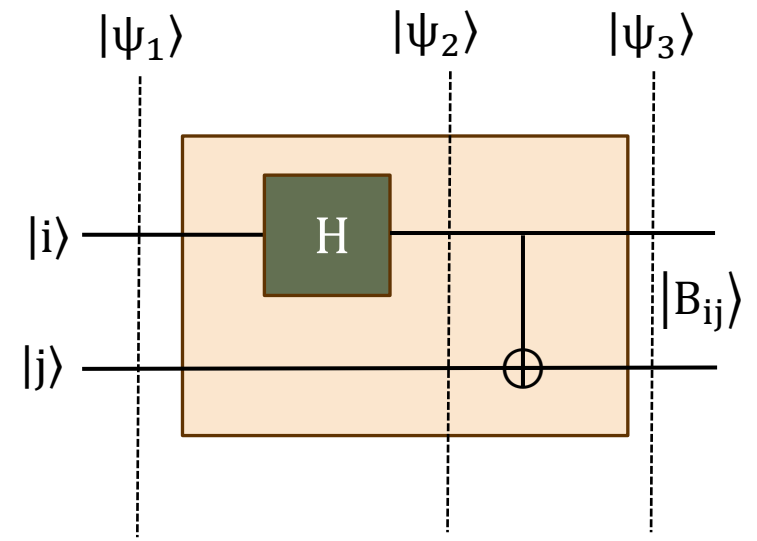
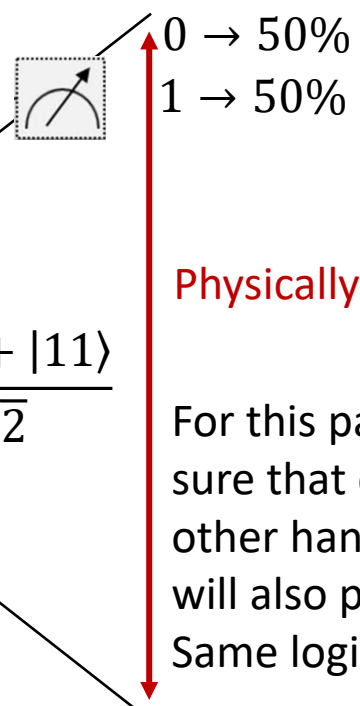
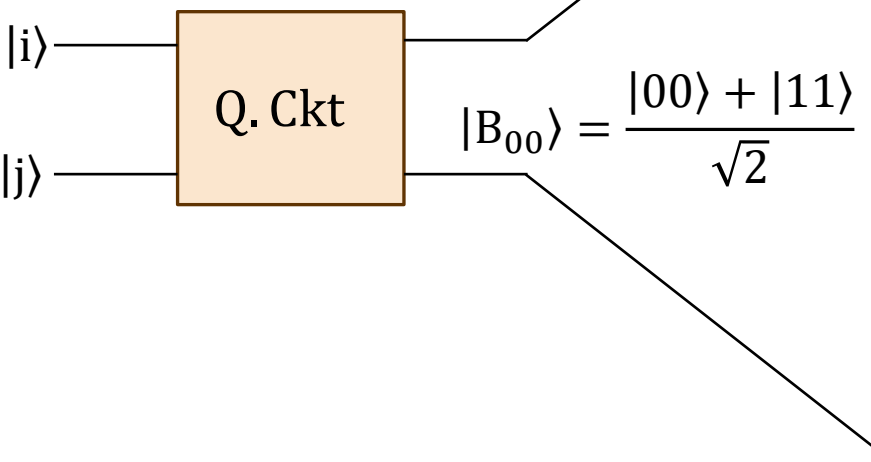
$$\left. \begin{aligned} |\phi^+\rangle &= |B_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \\ |\phi^-\rangle &= |B_{10}\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \\ |\psi^+\rangle &= |B_{01}\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} \\ |\psi^-\rangle &= |B_{11}\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} \end{aligned} \right\} \text{ Bell states}$$

- Smallest system where quantum correlations beat classical ones
- First place where “quantum advantage” appears
- If a device can’t create Bell states reliably, it’s not truly quantum

Entanglement

$$|\phi^+\rangle = |B_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$
$$|\phi^-\rangle = |B_{10}\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$
$$|\psi^+\rangle = |B_{01}\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$
$$|\psi^-\rangle = |B_{11}\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

Bell states



$|i\rangle = |1\rangle \quad |j\rangle = |0\rangle$
$$|\psi_1\rangle = |10\rangle$$
$$|\psi_2\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} |0\rangle = \frac{|00\rangle - |10\rangle}{\sqrt{2}}$$
$$|\psi_3\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |B_{10}\rangle = |\phi^-\rangle$$

Physically Separated

For this particular Bell state, if we measure qubit 1 to be '0', then we know for sure that qubit 2 will also produce '0' even without any measurement. On the other hand, if we measure qubit 1 to be '1', then we know for sure that qubit 2 will also produce '1' even without any measurement. Same logic, if instead of measuring qubit 1, we measure qubit 2.

Entanglement

Let us use Qiskit to create $|\psi^+\rangle = |B_{01}\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$

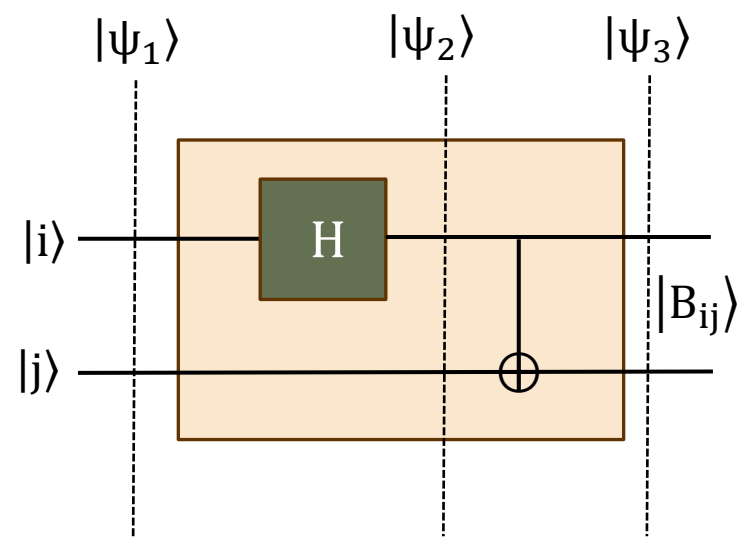
Start with two qubits in the state $|01\rangle$

Apply a Hadamard gate on qubit 0

Apply a CNOT gate with qubit 0 as control and qubit 1 as target

Measure both qubits

Show histogram



Use Qiskit to create an entangled state of three qubits.

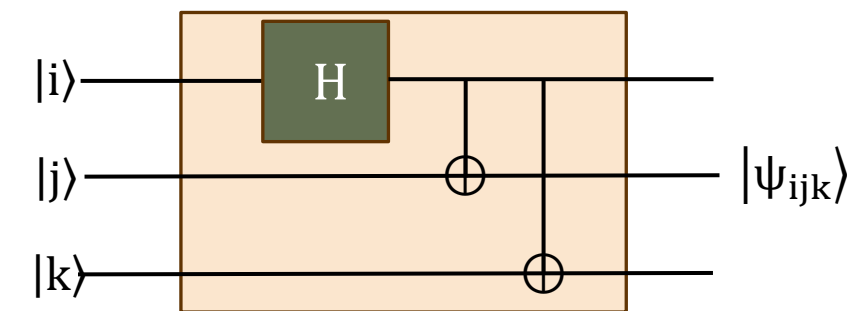
Measure it, and show that:

Each qubit individually looks random

The three qubits are perfectly correlated together

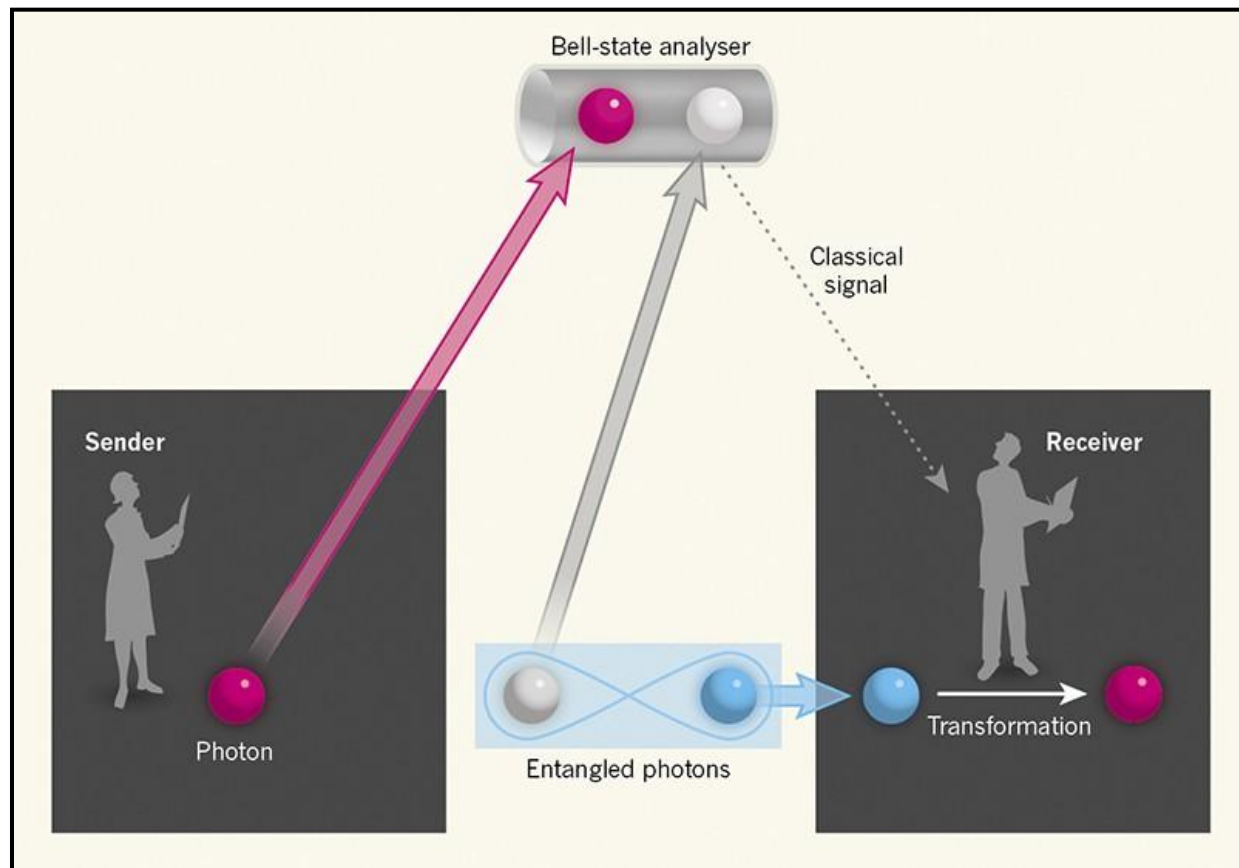
Bell states are two-qubit states and the simplest form of entangled states. This involves three qubits. Multi-qubit entangled states have different names. This one is called a GHZ state.

- Quantum error correction: $\rightarrow$ information spread across many qubits
- Quantum algorithms $\rightarrow$ speed-up needs multi-qubit entanglement
- Quantum networks $\rightarrow$ sharing information among many parties



```
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator
from qiskit.quantum_info import Statevector
from qiskit.visualization import plot_histogram
from qiskit.visualization import plot_bloch_multivector
```

Entanglement Application – Quantum Teleportation



Alice and Bob are far apart and want to transfer a quantum state from Alice to Bob. Alice does not know what is the qubit. She cannot measure it. Because of the no-cloning theorem, Alice cannot copy the qubit and send it. If she tries to copy or measure it, the state is destroyed. So, what do they do?

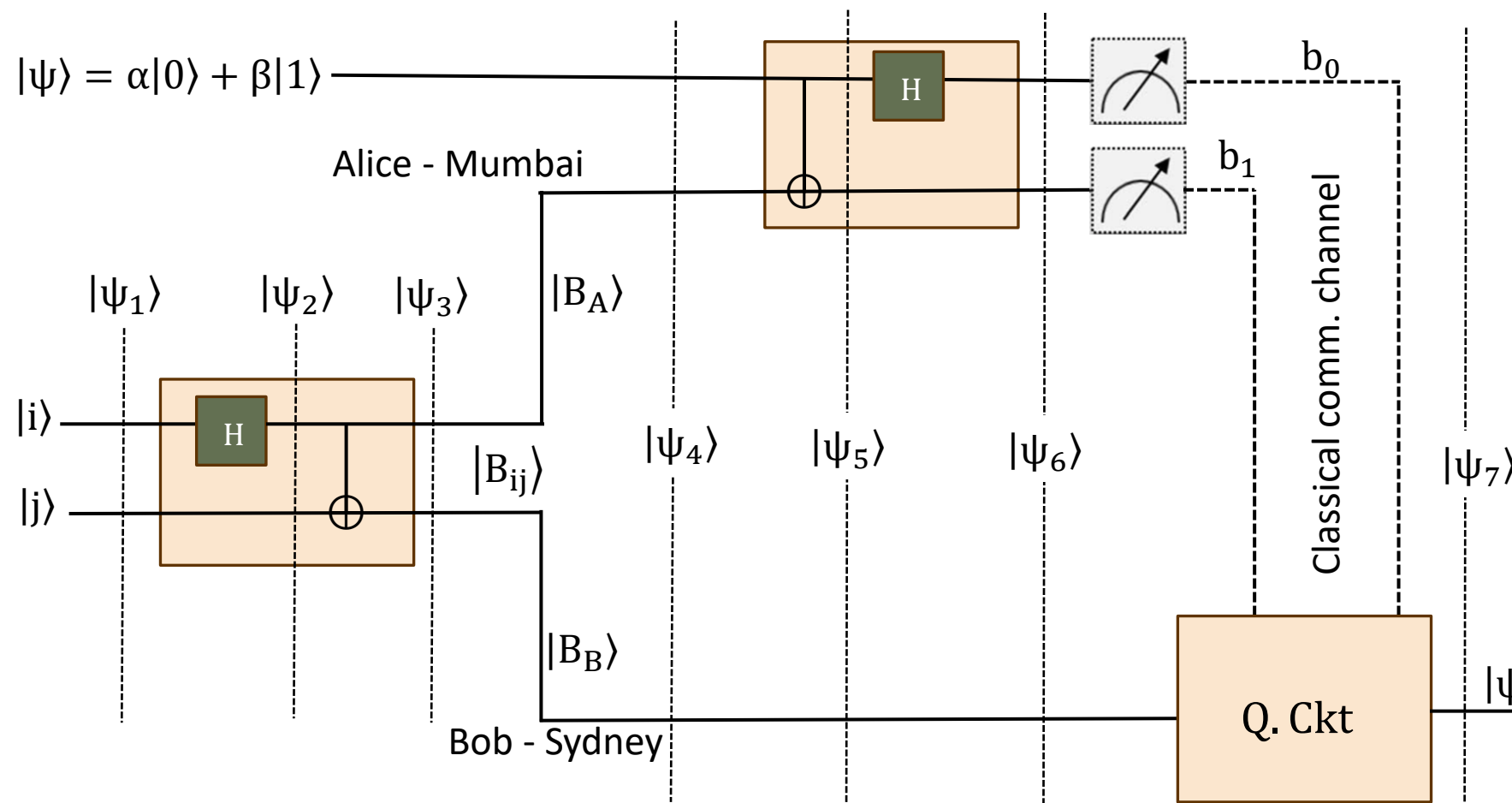
Before separating, Alice and Bob share a pair of entangled qubits (a Bell state). Alice keeps one qubit, Bob keeps the other. Now Alice performs a joint measurement on:

- her unknown qubit, and
- her half of the entangled pair.

This measurement does not reveal the quantum state, but it produces two classical bits. Alice sends those two classical bits to Bob using a normal (classical) channel. When Bob receives the bits, he performs a simple correction on his qubit.

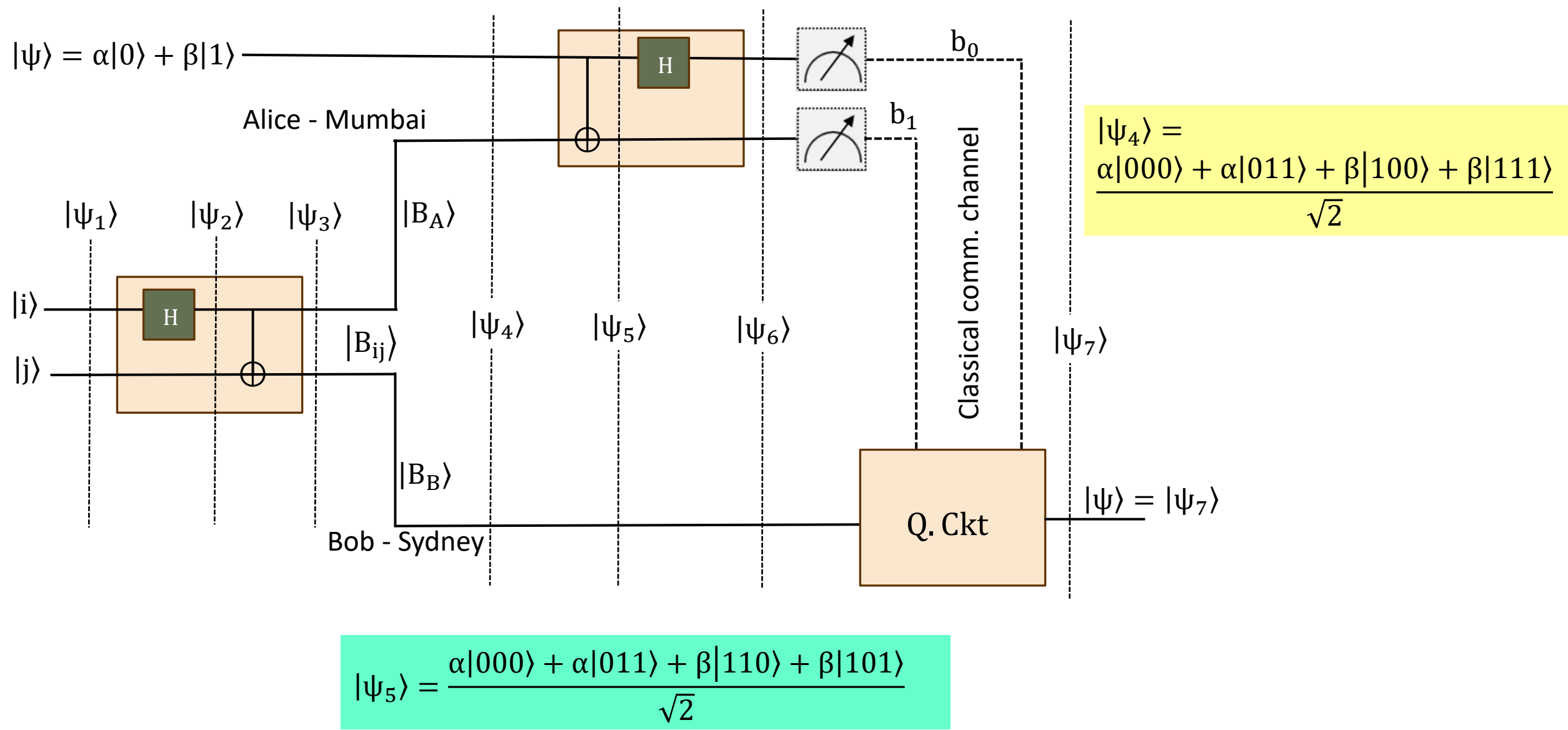
Magically, Bob's qubit becomes exactly the same quantum state that Alice had — even though the original state never traveled.

Entanglement Application – Quantum Teleportation

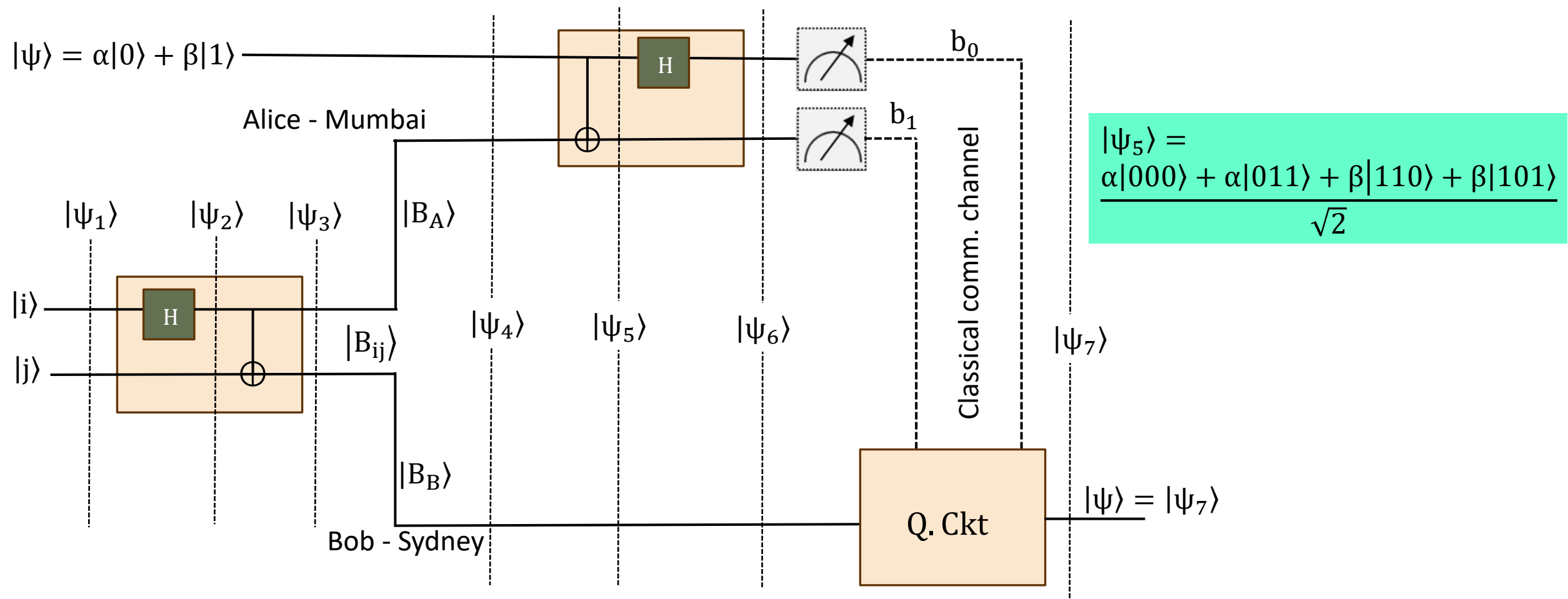


$$\begin{aligned} |\psi_1\rangle &= |00\rangle \\ |\psi_2\rangle &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |0\rangle \\ &= \frac{|00\rangle + |10\rangle}{\sqrt{2}} \\ |\psi_3\rangle &= \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |B_{00}\rangle \\ |\psi_4\rangle &= |\psi\rangle \otimes |B_{00}\rangle \\ &= (\alpha|0\rangle + \beta|1\rangle) \frac{|00\rangle + |11\rangle}{\sqrt{2}} \\ &= \frac{\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle}{\sqrt{2}} \end{aligned}$$

Entanglement Application – Quantum Teleportation



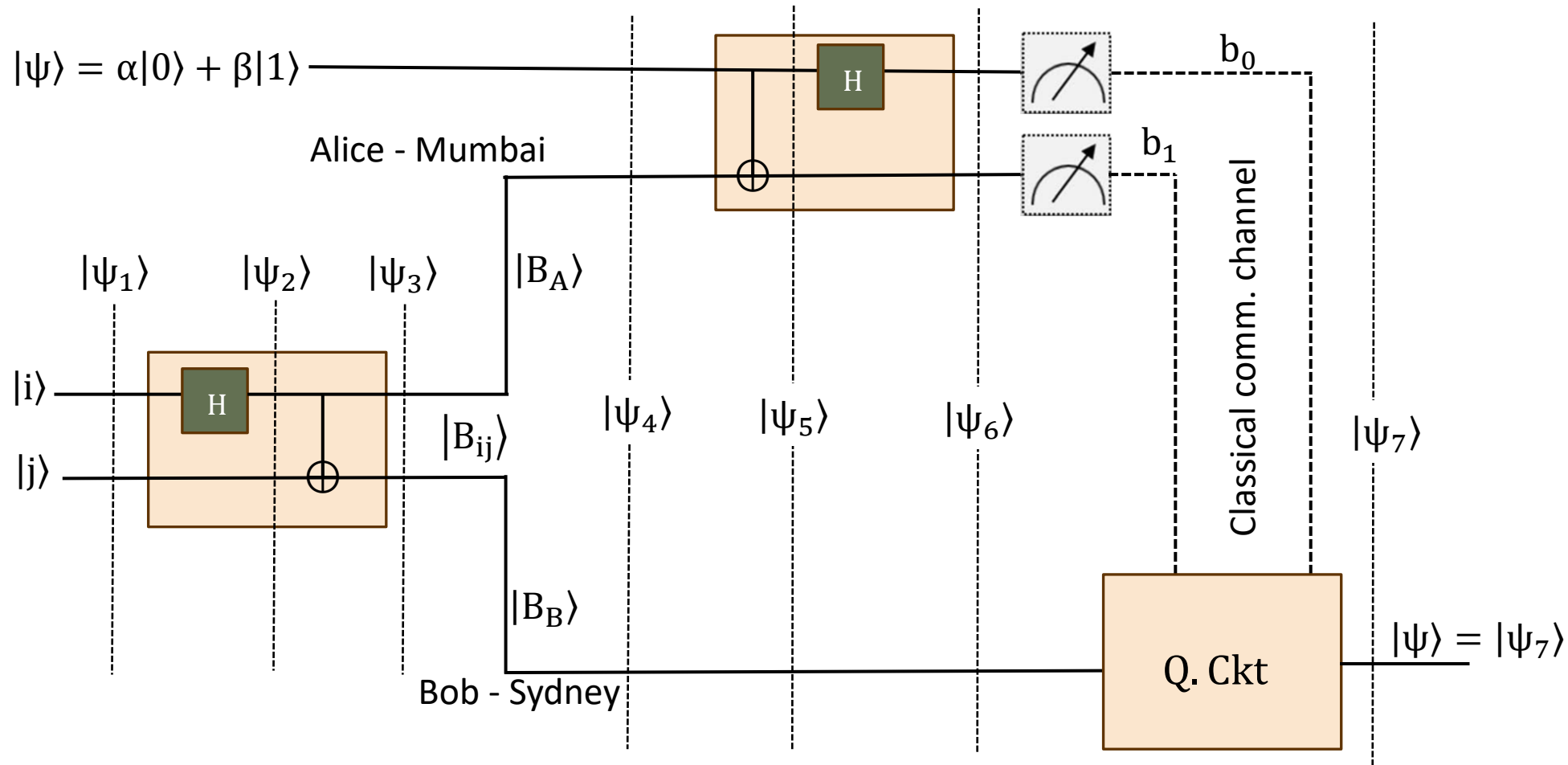
Entanglement Application – Quantum Teleportation



$$|\psi_5\rangle = \frac{\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle}{\sqrt{2}}$$

$$\begin{aligned} |\psi_6\rangle &= \frac{1}{\sqrt{2}} \left[\alpha \left\{ \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right\} |00\rangle + \alpha \left\{ \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right\} |11\rangle + \beta \left\{ \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\} |10\rangle + \beta \left\{ \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\} |01\rangle \right] \\ &= \frac{1}{2} [\alpha|000\rangle + \alpha|100\rangle + \alpha|011\rangle + \alpha|111\rangle + \beta|010\rangle - \beta|110\rangle + \beta|001\rangle - \beta|101\rangle] \end{aligned}$$

Entanglement Application – Quantum Teleportation



$$|\psi_6\rangle = \frac{1}{2} [\alpha|000\rangle + \alpha|100\rangle + \alpha|011\rangle + \alpha|111\rangle + \beta|010\rangle - \beta|110\rangle + \beta|001\rangle - \beta|101\rangle]$$
$$= \frac{1}{2} \left[\begin{matrix} \alpha|000\rangle + \alpha|100\rangle + \alpha|011\rangle + \alpha|111\rangle \\ + \beta|010\rangle - \beta|110\rangle + \beta|001\rangle - \beta|101\rangle \end{matrix} \right] = \frac{1}{2} \left[\begin{matrix} |00\rangle(\alpha|0\rangle + \beta|1\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) \\ + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle) \end{matrix} \right]$$

Entanglement Application – Quantum Teleportation

$$|\psi_6\rangle = \frac{1}{2} [|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)]$$

Measured bits by Alice (b_0, b_1)	Prob.	Resultant State at Bob's end	Operation needed by Bob $ \psi_7\rangle = \psi\rangle$
00	1/4	$\alpha 0\rangle + \beta 1\rangle$	Do nothing: $I B_B\rangle$
01	1/4	$\alpha 0\rangle - \beta 1\rangle$	$Z B_B\rangle$
10	1/4	$\alpha 1\rangle + \beta 0\rangle$	$X B_B\rangle$
11	1/4	$\alpha 1\rangle - \beta 0\rangle$	$Z\{X B_B\}$

Quantum teleportation does not move matter or energy. It moves information, using entanglement as the bridge

The measurement irreversibly destroys $|\psi\rangle$ at Alice's side. After that moment, $|\psi\rangle$ no longer exists anywhere in the universe. Only after Bob receives the classical bits and applies a correction does $|\psi\rangle$ reappear at Bob's side. At no point do two copies of $|\psi\rangle$ exist simultaneously – it does not violate the no-cloning theorem.

Entanglement Application – Quantum Teleportation

$|\phi^+\rangle$ or $|B_{00}\rangle$

Alice (b_0, b_1)	Bob applies
00	I
01	X
10	Z
11	ZX

$|\phi^-\rangle$ or $|B_{10}\rangle$

Alice (b_0, b_1)	Bob applies
00	Z
01	ZX
10	I
11	X

$|\psi^+\rangle = |B_{01}\rangle$

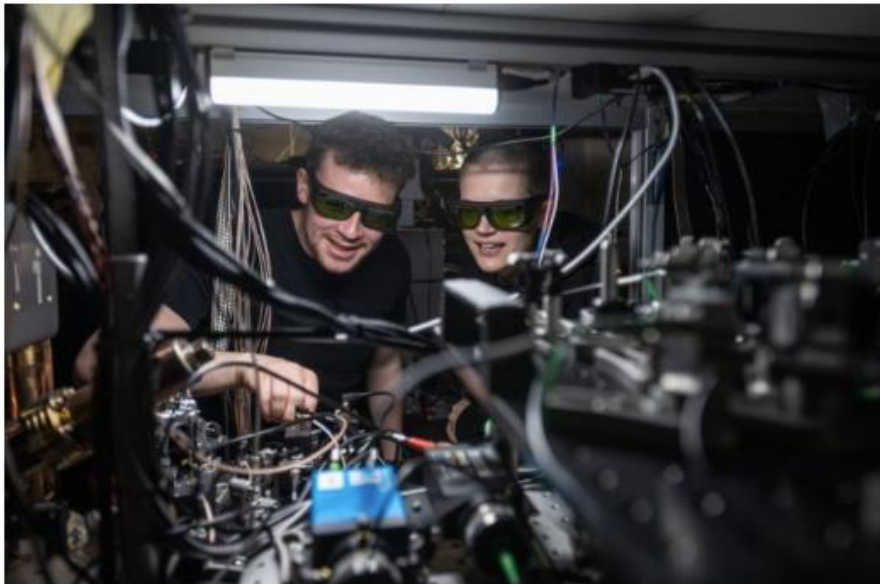
Alice (b_0, b_1)	Bob applies
00	X
01	I
10	ZX
11	Z

$|\psi^-\rangle$ or $|B_{11}\rangle$

Alice (b_0, b_1)	Bob applies
00	ZX
01	Z
10	X
11	I

Oxford quantum teleportation breakthrough brings scalable quantum computing closer to reality

Quantum / 10th February 2025



Wion / Photos / What Is Quantum Teleportation? How Scientists Made A Breakthrough For The First Time

What is Quantum Teleportation? How scientists made a breakthrough for the first time

Edited By **Vinay Prasad Sharma**
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Quantum teleportation sends information, not matter, using entanglement from one place to another. In 2024, scientists teleported a quantum state of light over 30 km of fibre-optic cable alongside live internet traffic.



Trending Photo



'India's homegrown AI era': 4 indigenous language models set to reshape the AI landscape



Lil Poppa dies at 25: Everything to know about the hip-hop artist's past life, girlfriend and relationship

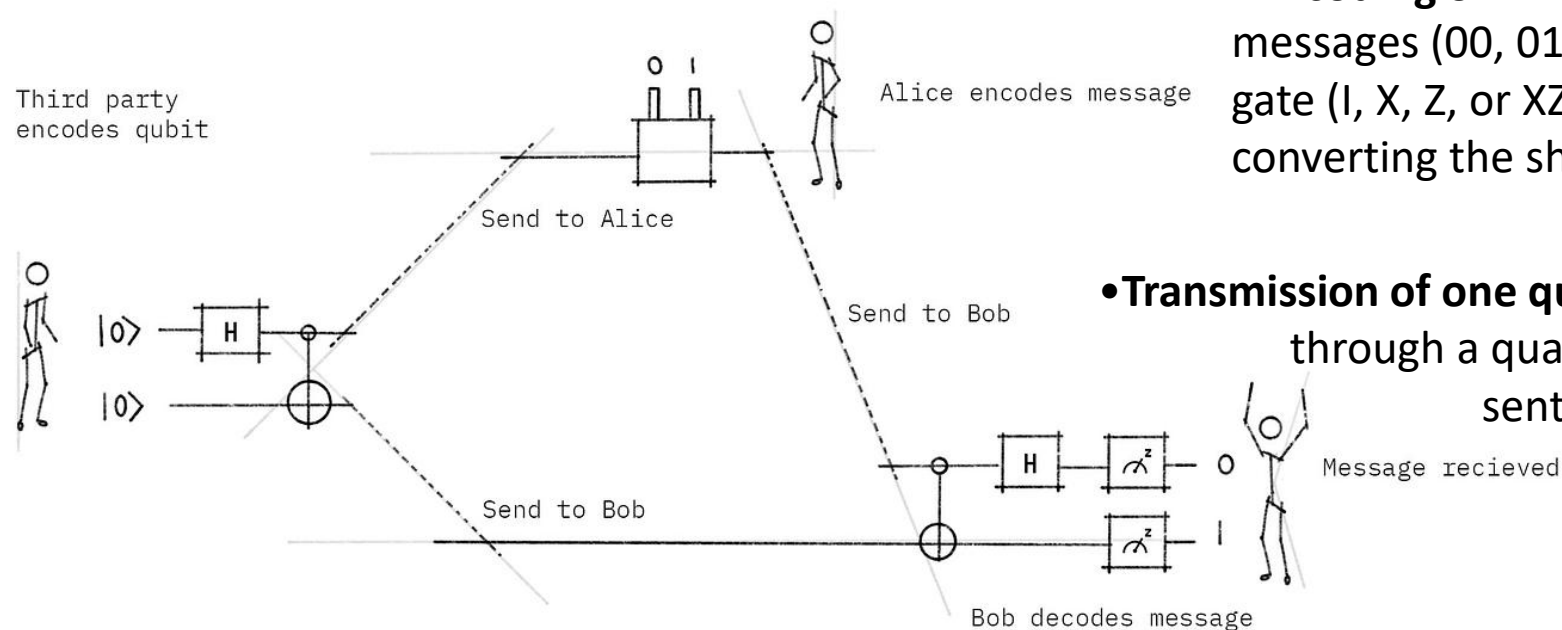


Entanglement Application – Superdense Coding

Alice and Bob live far apart and want to communicate.

Normally, if Alice sends one classical bit, Bob can learn one bit of information.

But quantum mechanics gives them a twist.



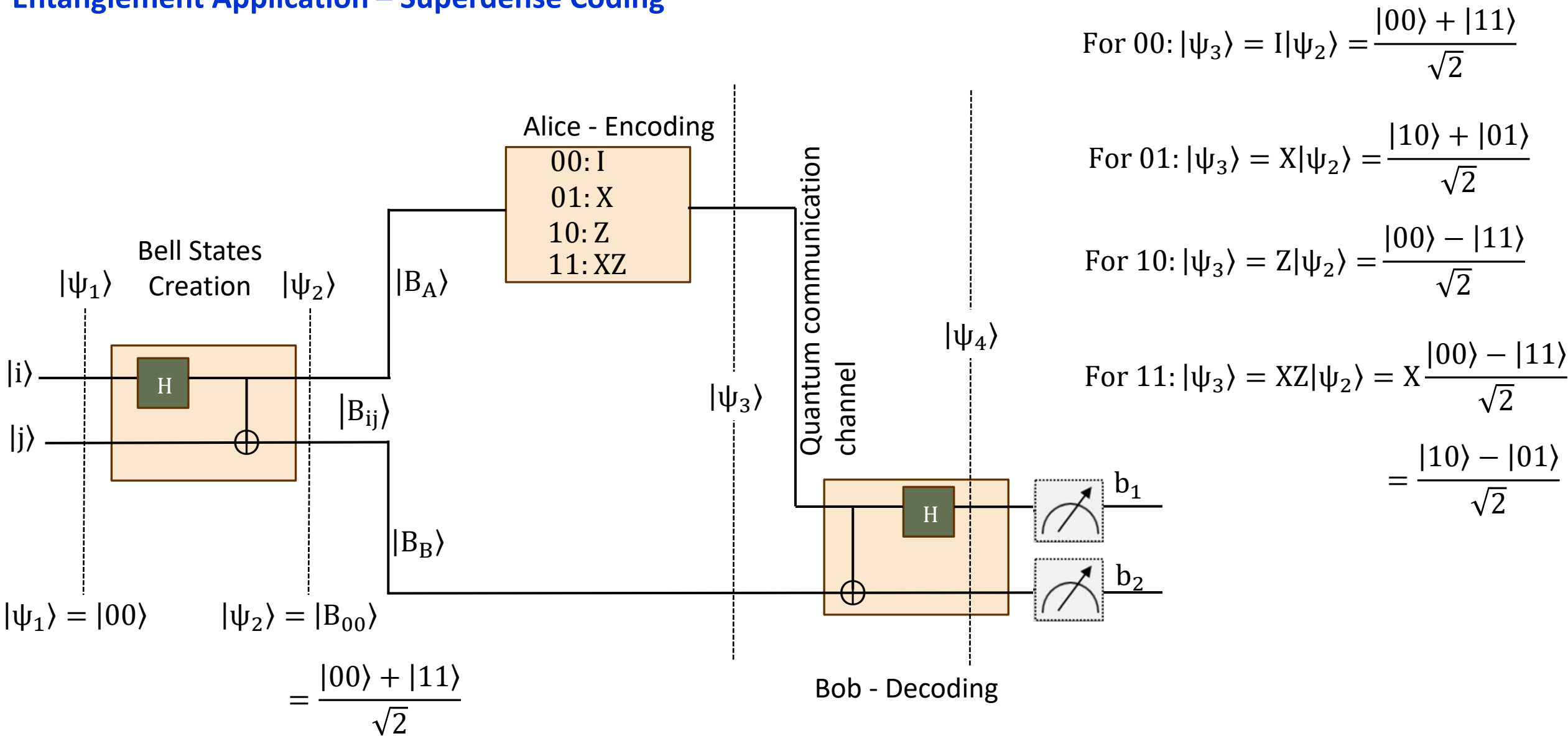
• **Encoding on Alice's qubit:** To send one of four classical messages (00, 01, 10, 11), Alice applies a chosen single-qubit gate (I, X, Z, or XZ) only on her half of the Bell pair, thereby converting the shared Bell state into a different Bell state.

• **Transmission of one qubit:** Alice then sends her qubit to Bob through a quantum channel. No classical information is sent at this stage. Only this single qubit travels.

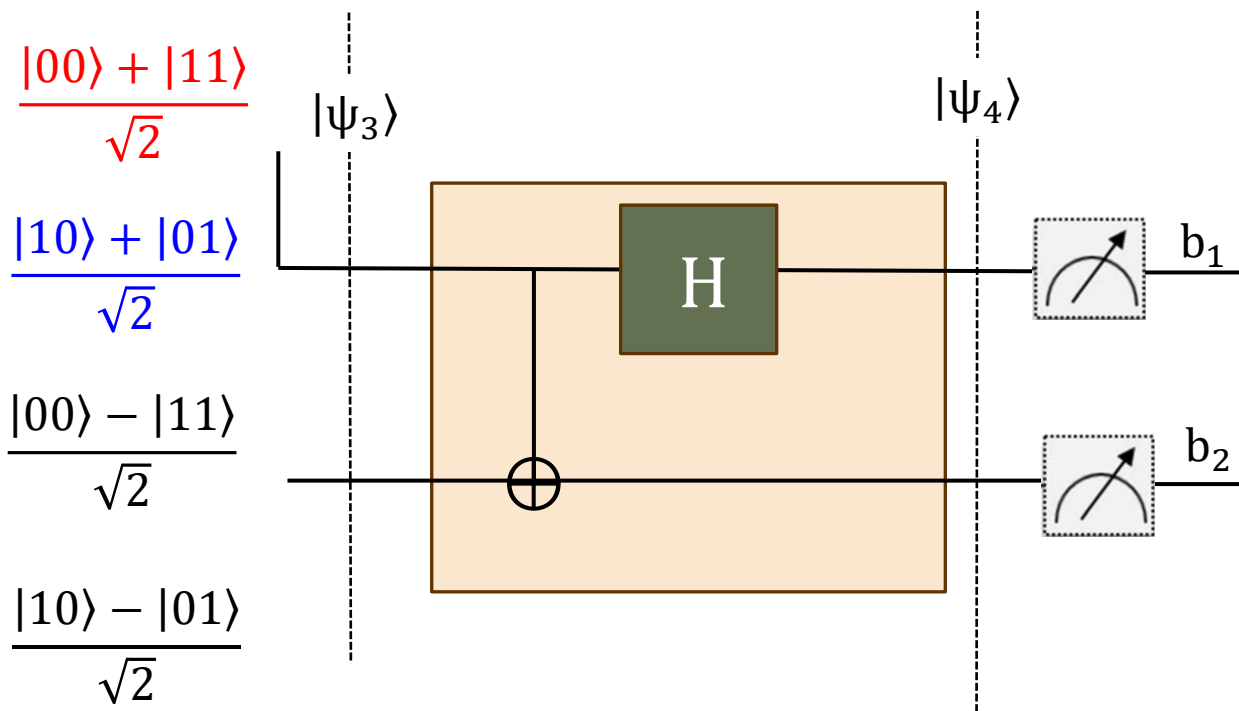
• **Shared Bell pair as a starting point:** Before communication begins, Alice and Bob share a specific Bell state (one qubit with Alice, the other with Bob), say the B_{00} state. This shared entangled pair is prepared in advance and kept intact.

• **Joint decoding at Bob:** Bob now possesses both qubits: his original half of the Bell pair and the qubit received from Alice. He performs a joint measurement on this two-qubit system (in the Bell basis) and uniquely identifies which Bell state they share, thereby recovering two classical bits.

Entanglement Application – Superdense Coding



Entanglement Application – Superdense Coding



$$\text{CNOT} \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

$$\begin{aligned} H \frac{|00\rangle + |10\rangle}{\sqrt{2}} &= \frac{1}{\sqrt{2}} \left[\frac{|0\rangle + |1\rangle}{\sqrt{2}} |0\rangle + \frac{|0\rangle - |1\rangle}{\sqrt{2}} |0\rangle \right] \\ &= \frac{1}{2} [|00\rangle + |10\rangle + |00\rangle - |10\rangle] = \frac{2|00\rangle}{2} = |00\rangle = |\psi_4\rangle \end{aligned}$$

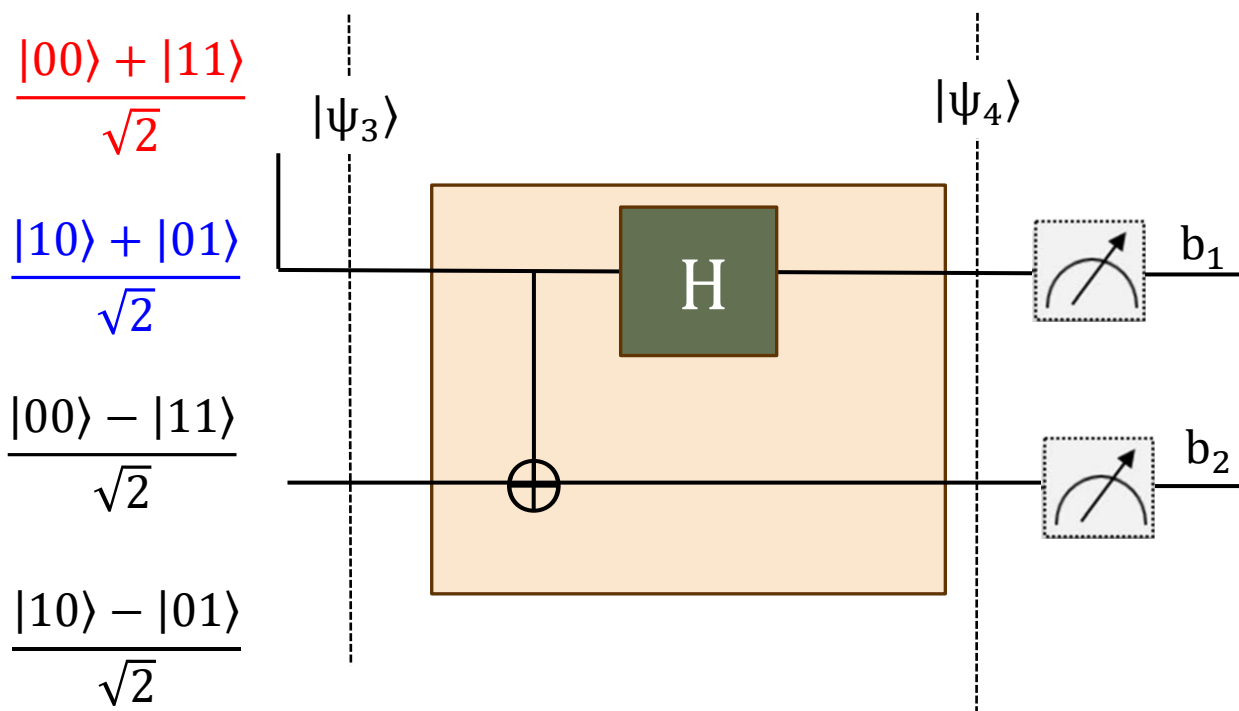
$$b_1 = 0; b_2 = 0$$

$$\text{CNOT} \frac{|10\rangle + |01\rangle}{\sqrt{2}} = \frac{|11\rangle + |01\rangle}{\sqrt{2}}$$

$$\begin{aligned} H \frac{|11\rangle + |01\rangle}{\sqrt{2}} &= \frac{1}{\sqrt{2}} \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} |1\rangle + \frac{|0\rangle + |1\rangle}{\sqrt{2}} |1\rangle \right] \\ &= \frac{1}{2} [|01\rangle - |11\rangle + |01\rangle + |11\rangle] = \frac{2|01\rangle}{2} = |01\rangle = |\psi_4\rangle \end{aligned}$$

$$b_1 = 0; b_2 = 1$$

Entanglement Application – Superdense Coding



The qubit Alice sends to Bob by itself carries no readable information. Only when Bob combines his portion of the Bell state with the qubit received from Alice and performs a joint measurement, only then can the two classical bits be recovered. The apparent doubling of capacity comes from using pre-shared entanglement as a communication resource, not from violating classical causality or information bounds.

There are exactly four Bell states, and Alice's four possible local operations map the shared pair one-to-one onto these states. Bob's measurement simply identifies which Bell state the two qubits are in, thereby decoding the message. Without Bell states, and the maximal entanglement they provide, superdense coding would not work.

- **Quantum communication efficiency:** Demonstrates how entanglement can increase the effective capacity of quantum channels.
- **Quantum networks:** Shows how pre-shared entanglement can be traded for reduced communication cost.
- **Foundations of quantum information:** Highlights that information can be stored and transmitted in correlations, not just physical carriers.


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Realizing ultrahigh capacity quantum superdense coding on quantum photonic chip



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Article

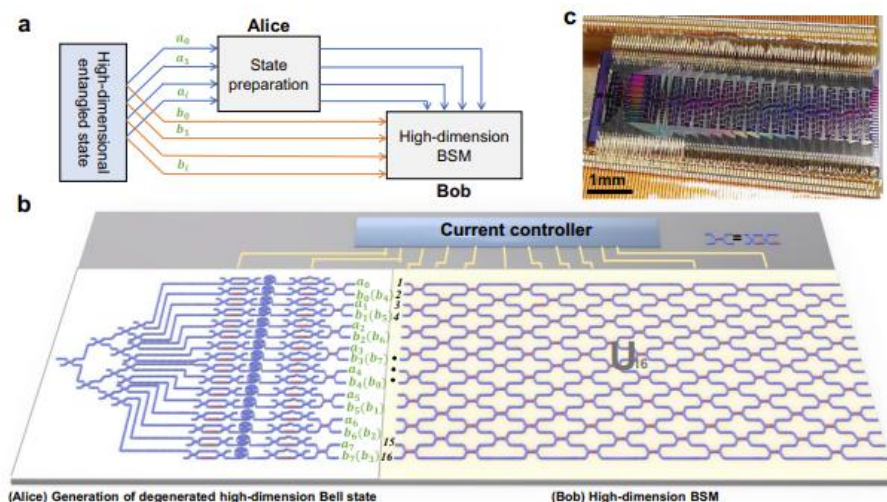


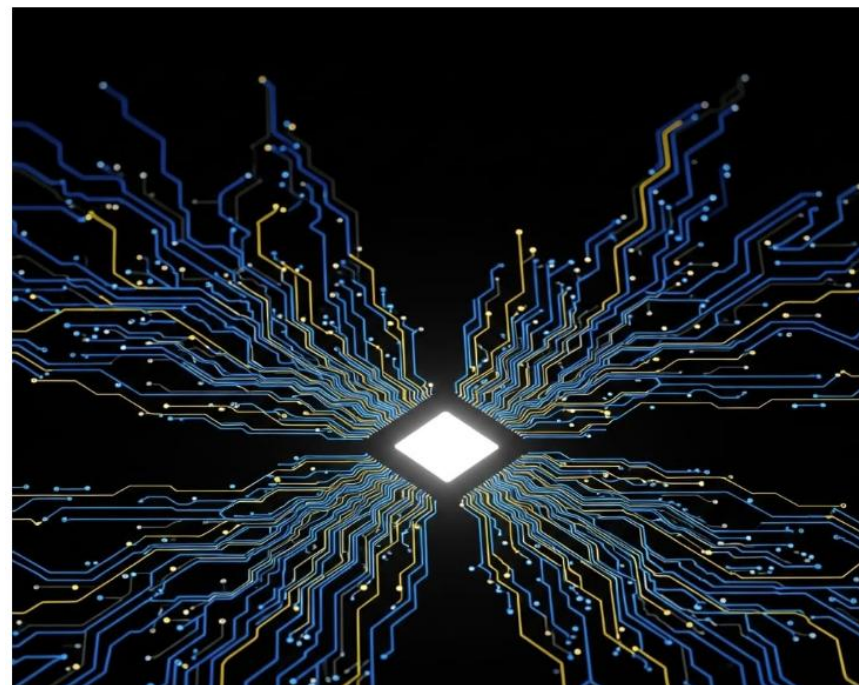
Fig. 1 | Quantum photonic chip for the realization of eight-dimensional quantum superdense coding. **a** The scheme of high-dimensional quantum superdense coding. **b** A dual-wavelength pump beam is injected into the photonic chip and separated into 16 paths to generate the weak single-mode squeezed states. Through the interference between the squeezed states, eight-dimensional Bell states for two photons (denoted as a and b) are generated. Subsequently, it is injected into a

16-mode unitary matrix which is composed of 120 MZIs for realizing the Bell state measurement. The heaters are precisely controlled by a current controller. The two photons are then injected into the superconducting nanowire single-photon detectors. Each MZI consists of two phase shifters and two multimode interferometers (MMI) **c** Photograph of the fabricated quantum photonic chip.

Entangled Light Boosts Data Capacity with New Coding Scheme



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BY ROHAIL T.



Scientists are continually seeking methods to enhance the capacity of communication channels, and superdense coding, a technique utilising entanglement, remains a pivotal area of investigation. Kai-Chi Chang from the Fang Lu Mesoscopic Optics and Quantum Electronics Laboratory at the University of California, Los Angeles, Arjun Mirani of the Department of Applied Physics at Stanford University, and Murat Can Sarihan, also from UCLA, alongside colleagues from the Departments of Physics at Stanford University and the Fang Lu Mesoscopic Optics and Quantum Electronics Laboratory at UCLA, present a novel high-dimensional superdense coding protocol employing energy-time entanglement. Their research details the creation of biphoton frequency combs, examples of entangled time-frequency Gottesman-Kitaev-Preskill states, and